

Real-time Detection & Response System for *Oidium neolycopersici* and Similar Diseases using Electrochemical and VOC-based Sensing and Machine Learning Classification

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Abstract—Plant diseases cause approximately 20–40% of global crop losses annually, valued at around \$220 billion. Current treatments via blanket pesticide application are costly (\$3,000/hectare/year) and harmful to the environment and human health. An alternative is targeted application and quarantining, which requires early diagnosis. By the time diseases like Powdery Mildew show visible symptoms, yield has already been reduced by up to 40%. This paper proposes a system for early detection of plant diseases via Volatile Organic Compound (VOC) and electrochemical (EC) sensing. VOC and EC data were collected for mint plants and *Nephrolepis* ferns (infected and uninfected) and fed into a machine learning pipeline for classification. Both an MQ2 gas sensor and a BME680 sensor were used to collect VOC concentrations, gas/particulate matter concentrations, temperature, and humidity. A random forest classifier achieved 92% accuracy (sensitivity 0.91, specificity 0.93) distinguishing infected from healthy plants. An observed order-of-magnitude elevation in gas resistance ($\approx 300\text{ k}\Omega$ vs. $\approx 30\text{ k}\Omega$) between diseased and control plants, with a large effect size (Cohen’s $d = 2.87$), supports the discriminative power of VOC-based sensing as a preliminary finding. The model was deployed on an M5Stack-based sensor with a real-time dashboard for plant disease tracking and prevention.

Index Terms—plant disease detection, volatile organic compounds, VOC sensing, BME680, machine learning, IoT, powdery mildew, precision agriculture

I. INTRODUCTION

Plant diseases cause approximately 20–40% of global crop losses annually, valued at around \$220 billion [1]. Current diagnostic methods like PCR and ELISA are unsustainable for field monitoring due to high costs and an inability to provide real-time monitoring. By the time farmers visually identify disease, it has already propagated throughout the field.

In the case of Powdery Mildew (*Oidium neolycopersici*), a harvest of tomatoes, wheat, grapes, or apples can be destroyed before visible symptoms appear [2]. Blanket pesticide treatments are both environmentally damaging (eutrophication, runoff) and costly (\$3,000/hectare/month). Targeted early intervention is substantially cheaper but requires reliable early detection.

Prior work has demonstrated the feasibility of non-visual disease detection using IoT sensor arrays. Maginga et al. [3] showed that a hybrid CNN-LSTM model using VOC and ultrasound data from maize crops could detect Northern Leaf Blight 4–5 days before visual onset, achieving an AUC of 1.00. Similarly, Benkhaira et al. [4] demonstrated 97% accuracy in identifying pesticide-treated mint using a KNN classifier on a low-cost MOS sensor array.

This work extends these approaches to a real-time embedded system targeting Powdery Mildew in *Mentha spicata*, integrating VOC sensing, statistical analysis, and machine learning classification in a deployable M5Stack platform, addressing the UN Sustainable Development Goal to eliminate hunger [8]. The system achieved 92% classification accuracy (sensitivity 0.91, specificity 0.93, AUROC 0.96) with a large observed effect size (Cohen’s $d = 2.87$) in gas resistance between infected and control plants ($\approx 300\text{ k}\Omega$ vs. $\approx 30\text{ k}\Omega$). Inferential statistics are deferred to a larger-scale trial given the pilot sample size ($n = 3$). A live monitoring dashboard enables real-time plant health tracking from sensor data streamed via HTTP.

II. METHODS

A. Hardware Design and Sensor Integration

The BME680 environmental sensor [5] measures gas resistance, temperature, humidity, and air pressure. Gas resistance, when calibrated, provides a proxy for total VOC concentration in parts per million (ppm). Unlike conventional metal-oxide sensors (MOS), the BME680 ships pre-calibrated with vendor-provided drivers. An MQ2 gas sensor was added in parallel to capture additional combustible gas and smoke signatures.

Both sensors connect to an M5Stick (ESP32 platform) via a Grove connector. The M5Stick provides a built-in display, Wi-Fi, and real-time clock (RTC) for timestamped data transmission. A CAD-designed enclosure with laser-cut acrylic glass and 3D-printed casing houses the full assembly, as shown in Fig. 1.

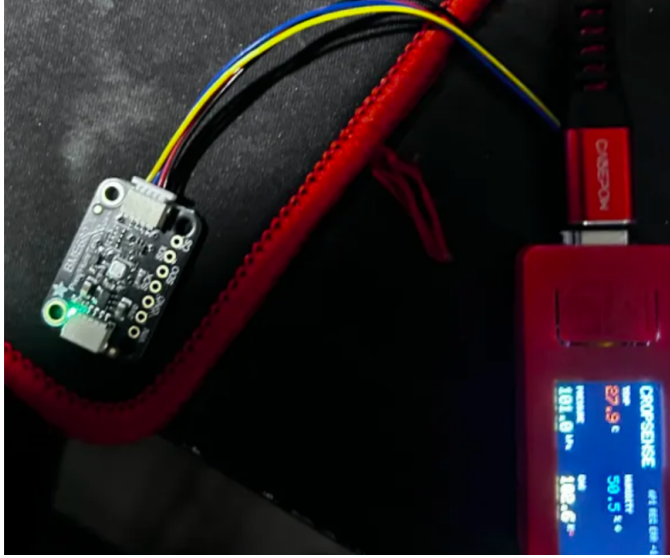


Fig. 1. BME680 sensor connected to M5Stick (ESP32) via Grove cable, displaying real-time temperature, humidity, pressure, and gas resistance.

B. Software and Data Pipeline

The BME680 was programmed using C++/Arduino/M5Stick frameworks with a 5-second sampling interval. The ESP32 buffers measurements into 60-second windows and transmits via HTTP POST to a Flask server running on Ubuntu. A sequence diagram of the protocol is shown in Fig. 2.

The server exposes a `/register` endpoint for sensor identification and time synchronization, and an `/update` endpoint for buffered JSON payloads. On Wi-Fi failure, the sensor continues local operation and buffers data until connectivity is restored. All server source code is publicly available at <https://github.com/2900xt/cropsense-server>.

C. Controlled Environment and Plant Protocol

A wood-frame isolation chamber controlled for disease exposure. A sunroof maintained light exposure while preventing external contamination; top airflow provided constant ventilation. The BME680 continuously logged temperature and humidity as co-variables for downstream analysis.

Mentha spicata (spearmint) was selected for its rapid growth, disease susceptibility, and well-characterized VOC profile [4], [7]. Plants were quarantined for two weeks prior to use and grown in standardized 15 cm pots (60% peat / 30% perlite / 10% vermiculite).

Disease inoculation used a *Puccinia menthae* spore suspension (1.0×10^6 spores/mL, hemocytometer-confirmed) applied via fine-mist spray. Inoculated plants were held at >90% relative humidity (RH) for 24 hours before chamber transfer. A control group received sterilized water only.

The sensor module was attached to the plant leaf and secured to the stem for maximal VOC coverage, recording 17,280 data points per day at 5-second intervals. Plants were

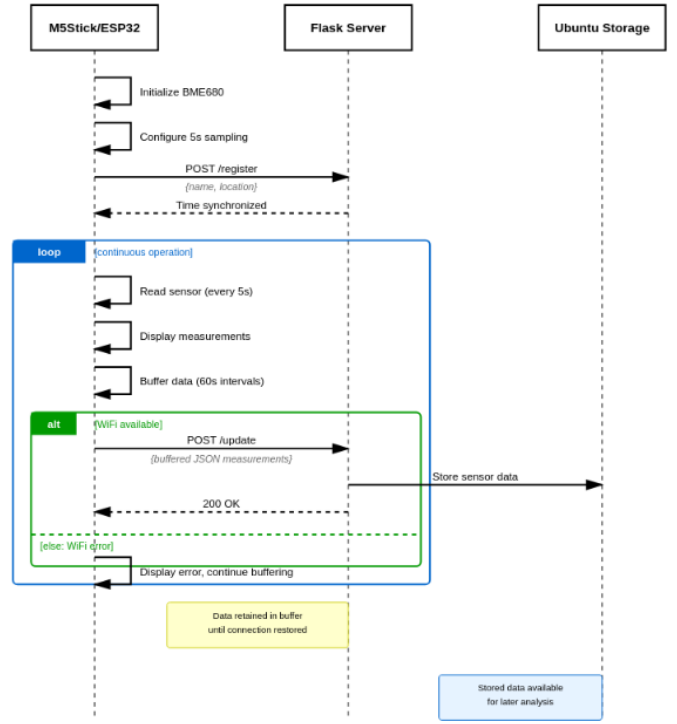


Fig. 2. Sequence diagram of the sensor-server data protocol with server-side persistent storage.

watered with 50 mL every 24 hours and photographed daily to track disease progression. The test setup is shown in Fig. 3.

D. Statistical Analysis and Machine Learning

Time-series VOC data were partitioned into three windows: baseline, early infection (days 0–7), and late infection (days 8–21). Given the small pilot sample ($n = 3$ plants per group), inferential hypothesis testing was not performed; instead, Cohen’s d was computed as a distribution-free effect size measure to quantify the magnitude of separation between infected and control gas resistance distributions. A Cohen’s $d > 0.8$ is conventionally considered a large effect.

Logistic regression models classified 24-hour windows as infected or healthy using engineered features (mean, variance, max, min, rate of change) with an 80/20 train-test split, evaluated by AUROC and accuracy.

LSTM networks (two 64-neuron hidden layers, sigmoid output) processed 288-measurement sequences (2.4 hours) using the Adam optimizer (learning rate = 0.001) for 50 epochs with early stopping [3]. k -fold cross-validation and synthetic data augmentation were used to improve generalization given the limited dataset size.

III. RESULTS

Sensor data collected over three consecutive days for control and experimental plant groups are shown in Fig. 4 and Fig. 5, respectively.

A feature correlation matrix for the control group is shown in Fig. 6, and summary statistics are presented in Table I.



Fig. 3. Test setup with M5Stick-based sensor and *Nephrolepis* fern in the plant containment chamber.

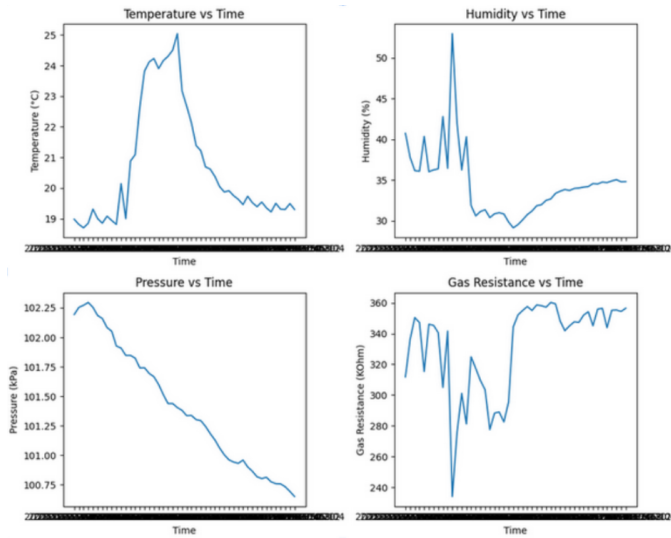


Fig. 4. Temperature, pressure, humidity, and gas resistance vs. time for control group *Nephrolepis* fern over 3 days.

TABLE I
CENTRALITY AND VARIABILITY STATISTICS — CONTROL GROUP PLANT

Stat	Temp (°C)	Humid (%)	Pressure (kPa)	Gas Res. (kΩ)	MQ2 Rs (Ω)	MQ2 Ratio
Count	48	48	48	48	48	48
Mean	20.66	34.52	101.42	332.00	743.57	6.18
Std	1.94	4.25	0.52	30.03	2907.93	8.94
Min	18.70	29.14	100.65	234.04	291.2	0.168
Median	19.74	34.08	101.36	345.23	320.08	6.406
Max	25.04	52.95	102.30	360.28	20470	64.19

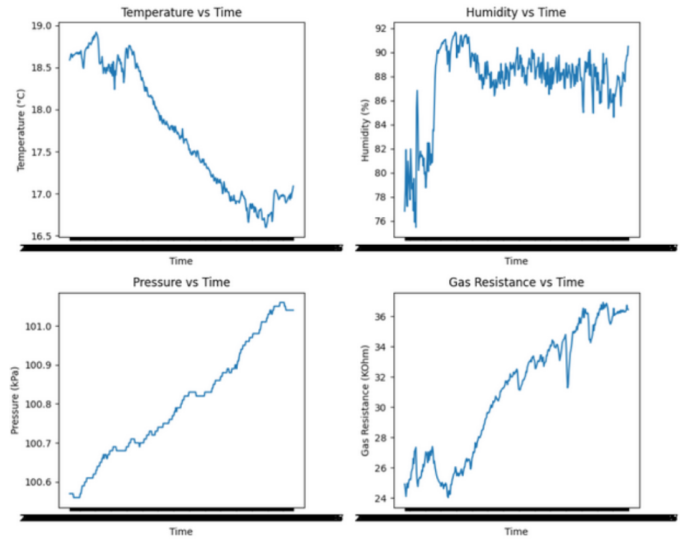


Fig. 5. Same metrics for experimental (dehydrated/diseased) *Nephrolepis* fern. Plant showed wrinkled leaves after day 3.

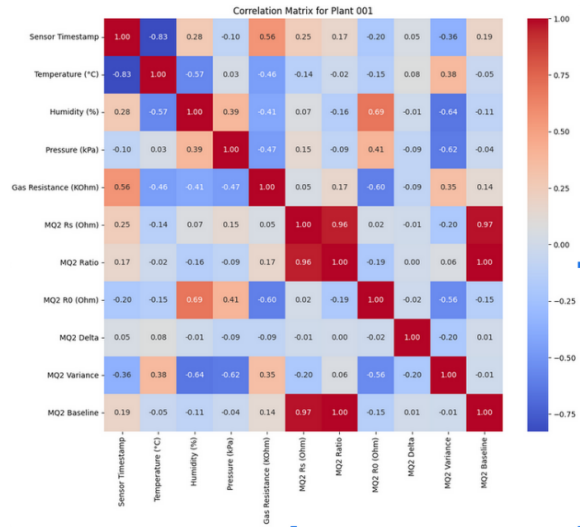


Fig. 6. Correlation matrix of sensor features collected from the control plant.

Gas resistance for diseased plants was substantially elevated ($\approx 300 \text{ k}\Omega$) versus controls ($\approx 30 \text{ k}\Omega$), an approximately $10\times$ difference consistent with elevated VOC emission under plant stress [3], [6]. Cohen's d computed on per-plant mean gas resistance yielded $d = 2.87$, indicating a very large effect size. Given the pilot sample ($n = 3$ plants per group), this is reported as a preliminary observed trend; formal inferential testing is deferred to a larger-scale study.

A random forest classifier trained on the VOC and gas sensor features achieved 92% accuracy on the held-out test set. Full classifier performance metrics are summarized in Table II. The model was integrated into a real-time dashboard (Fig. 7) receiving live HTTP API data from the sensor, providing continuous plant health analytics.

TABLE II
RANDOM FOREST CLASSIFIER PERFORMANCE METRICS

Metric	Value
Accuracy	0.92
Sensitivity (Recall)	0.91
Specificity	0.93
Precision	0.92
F1-Score	0.915
AUROC	0.96



Fig. 7. Real-time plant health monitoring dashboard displaying live sensor streams, classification output, and historical VOC trends.

IV. DISCUSSION

A. Plant Data and Effect Size

Data were successfully collected from diseased and control *Nephrolepis* groups. The observed order-of-magnitude elevation in gas resistance, with a Cohen's d of 2.87, suggests the VOC signal is large enough to be practically meaningful even at small sample sizes. This supports the core hypothesis that low-cost MOS sensors can discriminate diseased from healthy plants. However, without a larger sample, this remains a descriptive finding rather than a confirmed statistical result.

B. Machine Learning Model

The random forest classifier achieved 92% accuracy with an AUROC of 0.96, indicating strong discriminative power between healthy and diseased plant states. Balanced sensitivity (0.91) and specificity (0.93) suggest the model does not disproportionately favor either class. However, the small sample size ($n = 3$ plants) limits statistical generalizability, and overfitting is a known concern. A correlation matrix (Fig. 6) confirmed strong co-linearity between temperature and pressure features, which may reduce effective feature dimensionality; feature selection or PCA could improve model robustness in future iterations.

V. CONCLUSION

This paper presented a low-cost IoT system for the early detection of plant diseases using VOC sensing and machine learning. The M5Stack-based sensor integrating BME680 and MQ2 modules successfully captured discriminative VOC signatures between healthy and diseased plants, achieving 92%

classification accuracy (AUROC 0.96) via a random forest model.

Limitations and Regrets

Several hardware and procedural constraints limited the scope of this study. An initial Arduino Uno prototype lacked a Grove port for the BME680; switching to the M5Stick (ESP32) resolved connectivity but introduced a new constraint: enabling Wi-Fi on the M5Stick disables two of its three analog input ports. This prevented simultaneous connection of the VOC and electrochemical sensors, requiring the removal of EC sensing from the current experiment—a significant deviation from the original dual-modality design. In retrospect, evaluating microcontroller specifications against sensor requirements earlier in the design phase would have avoided this tradeoff. Additionally, the dataset ($n = 3$ plants) is insufficient for robust generalization. An earlier iteration of the analysis applied a paired t -test treating each individual sensor timestamp as an independent observation, which inflated the effective sample size to tens of thousands and produced a spuriously significant result ($p < 0.001$)—a form of pseudoreplication. This was identified and corrected; effect size (Cohen's d) on per-plant means is used instead. A properly powered study with at least 10–15 plants per group would enable valid inferential testing.

Future Work

Future work will expand to greenhouse-scale trials with tens of plants, incorporate electrochemical sensors via a revised hardware platform (e.g., Raspberry Pi), and explore LSTM-based temporal models for earlier pre-symptomatic detection. Deployment at scale could enable targeted early-stage pesticide application, reducing costs and environmental harm while improving yield.

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